

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

GLY 314 — Past Questions & Materials
by Renance

Course: GLY 314
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 300

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Paper 1 of 3 — GLY 314(3CU)

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF SCIENCE



DEPARTMENT OF GEOLOGY
FIRST SEMESTER EXAMINATION 2024/2025 ACADEMIC SESSION.
MINERALOGY, GLY 314 (3CU).

Time Allowed: 2½ Hrs

INSTRUCTION: ANSWER ANY FOUR QUESTIONS IN ALL.

SECTION A

Answer any two (2) Questions in this Section

1. (a) Define what a mineral is and explain all five essential characteristics that a substance must possess to be classified as a mineral. (4 marks)
 (b) Describe what causes magnetism in minerals and explain the difference between ferromagnetic and ferrimagnetic minerals. (5 marks)
 (c) Explain the concept of crystal habit. How does the environment in which a crystal grows affect its habit? Provide three examples of different crystal habits and minerals that might exhibit them. (8 ½ marks)
2. (a) Define polymorphism in the context of mineralogy and explain how changes in temperature and pressure can lead to polymorphic transformations. (6½ marks)
 (b) Describe the three types of polymorphic transformations (reconstructive, displacive, and order-disorder). Using at least two mineral examples, explain how these transformations influence the occurrence and stability of mineral polymorphs in different geological environments (11 marks)
3. (a) What are the primary differences between single-chain and double-chain silicates? Name one mineral group associated with each structure. (5 marks)
 (b) Compare and contrast the structure and properties of phyllosilicates and tectosilicates, explaining how their atomic arrangements influence their physical properties. Discuss how these structural differences affect their roles in crustal formation and geological processes. (12½ marks)

Section B

Answer any two (2) Questions in this Section.

4. (a) Enumerate the FOUR basic factors a mineralogist/petrologist must consider before carrying out any type of chemical analysis (6 marks)
 (b) Name and explain the various divisions of chemical analysis (5 ½marks)
 (c) Give a general overview of spectroscopic techniques (6 marks)
5. (a) Give a detailed classification of clay minerals by Bailey, 1980b (8½ marks)
 (b) List SIX properties of clay and discuss THREE (9 marks)
6. (a) Give a detailed account of Electron Microprobe and scanning electron microscopy (11½ marks)
 (b) In a tabular form, list similarities and differences in the operation of Energy Dispersive Spectrometers (EDS) and Wavelength Dispersive Spectrometers (WDS). (6 marks)



Paper 2 of 3 — MINERALOGY GLY 314 2024I2025

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF SCIENCE



DEPARTMENT OF GEOLOGY
FIRST SEMESTER EXAMINATION 2024/2025 ACADEMIC SESSION.
MINERALOGY, GLY 314 (3CU).

INSTRUCTION: ANSWER ANY FOUR QUESTIONS IN ALL.

Time Allowed: 2½ Hrs

SECTION A

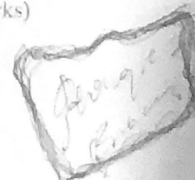
Answer any two (2) Questions in this Section

1. (a) Define what a mineral is and explain all five essential characteristics that a substance must possess to be classified as a mineral. (4 marks)
(b) Describe what causes magnetism in minerals and explain the difference between ferromagnetic and ferrimagnetic minerals. (5 marks)
(c) Explain the concept of crystal habit. How does the environment in which a crystal grows affect its habit? Provide three examples of different crystal habits and minerals that might exhibit them. (8 ½ marks)
2. (a) Define polymorphism in the context of mineralogy and explain how changes in temperature and pressure can lead to polymorphic transformations. (6½ marks)
(b) Describe the three types of polymorphic transformations (reconstructive, displacive, and order-disorder). Using at least two mineral examples, explain how these transformations influence the occurrence and stability of mineral polymorphs in different geological environments (11 marks)
3. (a) What are the primary differences between single-chain and double-chain silicates? Name one mineral group associated with each structure. (5 marks)
(b) Compare and contrast the structure and properties of phyllosilicates and tectosilicates, explaining how their atomic arrangements influence their physical properties. Discuss how these structural differences affect their roles in crustal formation and geological processes. (12½ marks)

Section B

Answer any two (2) Questions in this Section.

4. (a) Enumerate the FOUR basic factors a mineralogist/petrologist must consider before carrying out any type of chemical analysis (6 marks)
(b) Name and explain the various divisions of chemical analysis (5 ½marks)
(c) Give a general overview of spectroscopic techniques (6 marks)
5. (a) Give a detailed classification of clay minerals by Bailey, 1980b (8½ marks)
(b) List SIX properties of clay and discuss THREE (9 marks)
6. (a) Give a detailed account of Electron Microprobe and scanning electron microscopy (11½ marks)
(b) In a tabular form, list similarities and differences in the operation of Energy Dispersive Spectrometers (EDS) and Wavelength Dispersive Spectrometers (WDS). (6 marks)



Paper 3 of 3 — GLY 314 STRUCTURAL GEOLOGY 2017-2020 Past Questions.

Level 300 • First Semester • PastQuestion

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DEPARTMENT OF EARTH SCIENCES

FIRST SEMESTER: BSc (GEOLOGY) EXAMINATION 2019/2020 SESSION

COURSE: ***GLY314/GLY316 STRUCTURAL GEOLOGY

TIME ALLOWED: 2 Hours

INSTRUCTION: ANSWER FOUR QUESTIONS IN ALL.

Answer question ONE and any other question in Section A, and answer ALL questions in Section B.

Use well labelled and titled diagram(s) to illustrate your answer(s) where necessary.

SECTION A

1. A mineralized schist contact contains a pegmatite dike that trends N60°W. The foliation plane of the mineralized schist has an attitude of N10°E/80°NW.

- (a) Plot the above information using β -diagram. (5 marks)
 (b) What is the apparent dip of the foliation plane along the pegmatite dike? (2 marks)
 (c) What is the bearing and plunge of the pegmatite dike? (5 marks)
 (d) What is the rake angle of the pegmatite in the foliation plane of the schist? (4 marks)

2. Distinguish between the following pairs and support your answer with diagrams (15 marks)

- (i) Homogeneous and Heterogeneous strains
 (ii) Hydrostatic pressure and hydrostatic tension
 (iii) Heave and Throw
 (iv) Triaxial stress test and Uniaxial stress test.
 (v) Foliation and slickenside

3. Briefly describe the stress patterns that are associated with the following types of fault. (15 marks)

- (i) Normal fault
 (ii) Thrust fault
 (iii) Strike-slip fault

SECTION B

4. Given that the Young Modulus of a granitic pluton is $1.38 \times 10^{10} \text{ lb/m}^2$, and on deformation it gives a change in length of $1.8 \times 10^{-2} \text{ m}$. If the original length of the pluton is $3.62 \times 10^8 \text{ m}$. (17½ marks)

- (i) Calculate the stress the rock is subjected to. $\rightarrow 6.858 \times 10^{-1}$
 (ii) Calculate the change in length of a rock given that it has a Poisson's ratio of 0.32, an original diameter of $1.68 \times 10^8 \text{ m}$ and the diameter on extension is given as $1.84 \times 10^8 \text{ m}$, with an original length of $3.24 \times 10^8 \text{ m}$. $\rightarrow 1.1099 \times 10^9$
 (iii) Calculate the coefficient of rigidity if the Young Modulus of the rock is $1.82 \times 10^8 \text{ lb/m}^2$. $\rightarrow 6$

5. Discuss the phenomenon of Folding in rocks under the following. (17½ marks)

- (i) Attitude of beds
 (ii) Parts of a fold
 (iii) Nomenclature of folds